

THE ZERO PARADOX

ZP-H Addendum

The Snap Floor in Native Categories

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ZP-H derives the Binary Snap across four mathematical frameworks and locates the snap floor $\perp$ as the categorical bottom in each. In the main document that location is carried by three depth-index proxy categories — copies of $\mathbb{N}$ with the order $\leq$ (Q_2 BallDepth, InfoDepth, HilbDimDepth), engineered so that depth 0 is the initial object. This addendum does the stronger thing: it drops the snap floor into each framework's own category, taken directly from Mathlib, where the floor appears in that field's native syntax. A topologist, a functional analyst, and a probabilist each meet the same bottom, each in their own category and their own language.

Three genuine Lean functors realize the snap chain inside the standard categories: F_B into TopCat (topological spaces), F_D into ModuleCat $\mathbb{C}$ ($\mathbb{C}$ -modules), and F_C into the Kleisli category of the probability monad PMF (stochastic maps). Each is sorry-free, and each carries the snap floor to its category's categorical bottom — an inverse limit in TopCat, an initial object in ModuleCat $\mathbb{C}$ and in the Kleisli category. The bundled witness is `mc1_correspondence` (MC1Bridge.lean).

Two honesty boundaries hold throughout. The realized claim is correspondence, not identity. What is proved is that each bottom is its own category's categorical bottom, agreeing on the snap; that the four bottoms are numerically one object is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence); what separates the members is proved property by property (`seam_unique_among_named`, for the named bottoms, in a lattice with no top). The native categories are the honest target. Mathlib has no bespoke category of p -adic spaces, Hilbert spaces, or information spaces, so the general-purpose categories — topological spaces, $\mathbb{C}$ -modules, stochastic maps — are where the floor genuinely lives.

Section I: From Depth Proxies to Native Categories

The proxy categories of the main document are correct but indirect. Q_2 BallDepth, InfoDepth, and HilbDimDepth are each a copy of $\mathbb{N}$ with the order $\leq$, wrapped so that the bottom 0 satisfies the universal property of an initial object and the source-asymmetry condition AX-G2 (no morphism returns to 0). They witness the snap floor's categorical role, but inside a structure built for the purpose rather than inside the domain itself.

The native categories are not built for the purpose, and that is the point — they also do not satisfy the ZP categorical axioms. A ZPCategory (ZP-G) requires that no terminal object exist (AX-G1): the snap has nowhere higher to go. TopCat, ModuleCat $\mathbb{C}$ and the Kleisli category all have a terminal object (the one-point space; the zero module; the one-point type, on which the only distribution is the point mass — the

example after `kleisli_bottom_not_terminal` in `ZeroParadox/Category/TreeSeam.lean`). Of the categories on this page, `AX-G1` holds only in the three depth-proxy categories, where the depth index has no maximum. So the snap floor cannot be "the initial object of a `ZPCategory`" here. The honest statement is the one each category supports natively: in `TopCat` the floor is the limit of the shrinking system, and in `ModuleCat` $\mathbb{C}$ and the Kleisli category it is the genuine initial object. Each is stated below.

Section II: `F_B` — The Snap Floor in `TopCat`

The 2-adic snap floor is $0 \in \mathbb{Q}_2$. Around it sits the descending tower of clopen balls $B(0, 2^{-n})$ — the unit ball, then radius $1/2$, then $1/4$, and so on — each a genuine topological subspace of $\mathbb{Q}_2$. The inclusions $B(0, 2^{-m}) \subseteq B(0, 2^{-n})$ for $n \leq m$ are continuous, so the tower is a functor out of $\mathbb{N}^{\text{op}}$ (the balls shrink as the index grows): a real diagram in `TopCat`.

Functor: <code>fB_functor : $\mathbb{N}^{\text{op}} \rightarrow \text{TopCat}$ (TopFunctor.lean)</code>
Object $n \mapsto$ the clopen ball $B(0, 2^{-n}) \subseteq \mathbb{Q}_2$, as a <code>TopCat</code> object.
Morphism ($n \leq m$): the continuous inclusion of the smaller ball into the larger.
<code>fB_bottom_is_limit</code> : $\bigcap_n B(0, 2^{-n}) = \{0\}$
The snap floor is the inverse limit of the tower — the one point common to every ball.
Sorry-free. Purity [<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>].

`TopCat` has both a terminal object (the one-point space) and an initial object (the empty space). The snap floor $\{0\}$ is a one-point space, so it is terminal (the first example after `padic_bottom_not_initial` in `ZeroParadox/Multihomed/TreeObstructions.lean`) and not initial (`padic_bottom_not_initial`). It is also the limit of the ball tower — the categorical expression a topologist would expect for "the point everything converges to." This is the honest realization in a category that is not a `ZPCategory`: not an initial object, but a limit.

Section III: `F_D` — The Snap Floor in `ModuleCat` $\mathbb{C}$

The state layer represents the snap in a complex Hilbert space: `StateSpace n` is the n -dimensional space $\mathbb{C}^n$, with the snap an orthogonal shift between basis states (`ZP-D`). As $\mathbb{C}$ -modules, these spaces and the linear maps between them form a diagram in `ModuleCat` $\mathbb{C}$.

Functor: <code>fD_functor : $\mathbb{N} \rightarrow \text{ModuleCat } \mathbb{C}$ (HilbFunctor.lean)</code>
Object $n \mapsto \text{StateSpace } n =$ the $\mathbb{C}$ -module $\mathbb{C}^n$.
Morphism ($n \leq m$): the isometric $\mathbb{C}$ -linear embedding that pads the extra coordinates with zero.
<code>fD_zero_isInitial</code> : <code>StateSpace 0</code> is the initial object of <code>ModuleCat</code> $\mathbb{C}$.

Functor: $\text{fD_functor} : \mathbb{N} \rightarrow \text{ModuleCat } \mathbb{C}$ (HilbFunctor.lean)

The snap floor is the zero module — the one-element $\mathbb{C}$ -module $\{0\}$, which is the categorical zero object (both initial and terminal).

fD_embed_inner : the embeddings preserve the inner product (genuine isometries).

Sorry-free. Purity [propext, Classical.choice, Quot.sound].

Here, unlike TopCat, the floor is the initial object outright: the zero object of $\text{ModuleCat } \mathbb{C}$ is $\text{StateSpace } 0$. The isometry lemma records what $\text{ModuleCat } \mathbb{C}$ forgets — that the embeddings are not merely linear but inner-product preserving — so the Hilbert-space structure travels alongside the functor rather than being discarded by the choice of category.

Section IV: F_C — The Snap Floor in KleisliCat PMF

The information layer measures the snap as one bit of divergence between distributions (ZP-C, $\text{JSD} = \log 2$). Its native category is the Kleisli category of the finite-distribution monad PMF: objects are finite types, and a morphism $A \rightarrow B$ is a stochastic map (a Markov kernel), assigning to each $a \in A$ a probability distribution on B . Mathlib supplies this category for free, since PMF is a lawful monad.

Functor: $\text{fC_functor} : \mathbb{N} \rightarrow \text{KleisliCat PMF}$ (InfoFunctor.lean)

Object $n \mapsto \text{Fin } n$, the type of n distinguishable outcomes.

Morphism ($n \leq m$): the deterministic embedding (each outcome mapped to its image with certainty).

fC_zero_isInitial : $\text{Fin } 0$ (the empty type) is the initial object.

fC_no_return : for $n > 0$ there is NO stochastic map $\text{Fin } n \rightarrow \text{Fin } 0$.

AX-G2 as a theorem: no probability distribution lives on the empty type, so nothing returns to the floor.

Sorry-free. Purity [propext, Classical.choice, Quot.sound].

The empty type is the snap floor here, and it carries the snap's irreversibility for free. A stochastic map into $\text{Fin } 0$ would be a probability distribution on the empty type, and there is none. So while every object has a unique map out of $\text{Fin } 0$ (it is initial), no nonempty object has any map back — AX-G2 (source asymmetry) is not assumed but proved, as fC_no_return . Of the three native categories, this is the one where the framework's irreversibility axiom becomes a theorem of the ambient mathematics.

Section V: The MC-1 Correspondence, Scope, and Purity

The three functors, together with F_A (the join-semilattice $\mathbb{N}$ of the main document, where 0 is already the initial object), are bundled as a single witness.

Definition: `mc1_correspondence` : `MC1Correspondence` (`MC1Bridge.lean`)

A record collecting the native-category realizations:

- `F_D`: `StateSpace 0` is initial in `ModuleCat C`.
- `F_C`: `Fin 0` is initial in `KleisliCat PMF`, with no return morphism.
- `F_B`: the clopen-ball tower has inverse limit $\{0\}$ in `TopCat`.

Sorry-free. Purity [`propext`, `Classical.choice`, `Quot.sound`].

This settles the correspondence half of MC-1: across the native categories, the snap floor is each category's own categorical bottom, and the four agree on the snap. The identity half, that the algebraic $\perp$, the 2-adic 0, the zero module, and the empty type are numerically one object, is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence); what separates the members is proved property by property (`seam_unique_among_named`, for the named bottoms, in a lattice with no top).

Scope and Purity

This is a formalisation milestone, not new category theory. `TopCat`, `ModuleCat C`, and the Kleisli category of `PMF` are standard Mathlib categories with standard universal properties. The contribution is realizing the ZP snap floor inside each as a genuine functor, in place of the $\mathbb{N}$ -indexed depth proxies of the main document.

Purity: [`propext`, `Classical.choice`, `Quot.sound`]. Unlike the choice-free ZP-J results, these functors inherit `Classical.choice` from Mathlib's topology, inner-product, and `PMF` libraries. The dependency is a library inheritance, not a new commitment of the construction; whether it is structurally forced is the open question tracked under `Classical.choice necessity` (ZP-L / ZP-M).

Lean Source Files

`TopFunctor.lean` — `fB_functor`, `fB_bottom_is_limit` (`F_B` into `TopCat`).

`HilbFunctor.lean` — `fD_functor`, `fD_zero_isInitial`, `fD_embed_inner` (`F_D` into `ModuleCat C`).

`InfoFunctor.lean` — `fC_functor`, `fC_zero_isInitial`, `fC_no_return` (`F_C` into `KleisliCat PMF`).

`MC1Bridge.lean` — `mc1_correspondence` (the bundled witness).

All in `ZeroParadox/` in the public repository.

Endnote: This document is an addendum to ZP-H Categorical Bridge and reads after it. ZP-H locates the snap floor as the categorical bottom across four frameworks using depth-index proxy categories; this addendum realizes the same floor inside each framework's native Mathlib category — `TopCat`, `ModuleCat C`, and the Kleisli category of distributions.

All results sorry-free in Lean 4 as of June 2026, footprint [`propext`, `Classical.choice`, `Quot.sound`].